

Full Rewritten Report (Second Copy)

SECOND COPY: FULLY REWRITTEN REPORT

1. Introduction

This report examines the factors influencing salary using a comprehensive dataset containing age, gender, education level, job title, years of experience, and salary. The analysis addresses five primary research questions and builds a series of statistical models to understand salary determinants.

2. Data Preparation

Data cleaning included standardizing education levels, removing invalid entries, and creating an interaction term between age and experience. These steps ensured accuracy and improved model performance.

3. Descriptive Statistics

Salary, age, and experience exhibited meaningful variation across the dataset. Higher education levels and leadership roles showed clear upward salary trends, motivating regression-based modeling.

4. Regression Models and Findings

A. Experience-only Model

Experience strongly predicts salary, explaining 65% of observed variation with a marginal increase around \$7,046 per year.

B. Adding Age and Interaction

Introducing age and interaction terms improved model fit ($R^2 \approx 0.72$). Age alone was significant initially but became non-significant once the interaction term was included, indicating experience returns vary with age.

C. Gender Model

Gender showed statistically significant differences. Men earned about \$13,507 more than women in the simple model, narrowing to \$6,207 in the full model.

D. Job Title Model

Job title explained 66% of salary variation, ranking among the strongest predictors. Executive titles produced the highest positive salary deviations.

E. Full Interaction Model

The strongest model included age, gender, education, experience, and the interaction term ($R^2 \approx 0.751$). Experience dominated prediction, with education also contributing meaningfully.

5. Research Question Answers

1. Experience strongly increases salary, with between \$7,000–\$16,000 added per year depending on the model.
2. Age does not independently predict salary after accounting for experience; its influence operates through interaction.
3. Gender and job title both show notable salary differences, with job title producing the largest effects.
4. A strong predictive model exists, with the full interaction model offering the best performance.
5. Salary increases by approximately \$7k–\$16k per year of experience depending on controls applied.

6. Conclusion

Salary is driven by experience, education level, and job title, with gender playing a smaller role and age influencing salary indirectly. The full interaction model best explains salary patterns.